MAT2002 HW5 Solutions

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1. General Solution of the System

For the system $x' = Ax$ where $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix}$: First, we find the eigenvalues by solving the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ -8 & -5 & -3-\lambda \end{pmatrix} = 0 \tag{1}$$

Expanding the determinant yields $-(\lambda^3 + \lambda^2 - 4\lambda - 4) = 0$, which factors as $-(\lambda^2 - 4)(\lambda + 1) = 0$. The eigenvalues are $\lambda_1 = -1$, $\lambda_2 = 2$, and $\lambda_3 = -2$. Next, we find the corresponding eigenvectors:

- For $\lambda_1 = -1$: $(A + I)v_1 = 0 \implies \begin{pmatrix} 2 & 1 & 1 \\ 2 & 2 & -1 \\ -8 & -5 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.
- Row reduction gives $v_1 = \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$.
- For $\lambda_2 = 2$: $(A - 2I)v_2 = 0 \implies \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -8 & -5 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.
- Row reduction gives $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.
- For $\lambda_3 = -2$: $(A + 2I)v_3 = 0 \implies \begin{pmatrix} 3 & 1 & 1 \\ 2 & 3 & -1 \\ -8 & -5 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.
- Row reduction gives $v_3 = \begin{pmatrix} -4 \\ 5 \\ 7 \end{pmatrix}$.

The general solution is:

$$x(t) = c_1 e^{-t} \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + c_3 e^{-2t} \begin{pmatrix} -4 \\ 5 \\ 7 \end{pmatrix} \tag{2}$$

2. Initial Value Problems & Asymptotic Behavior

(i) For the system $x' = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix} x$ with $x(0) = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$: The characteristic polynomial is $-(\lambda-1)(\lambda-2)(\lambda-3) = 0$, giving eigenvalues $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$. Eigenvectors are found by finding the null space of $(A - \lambda I)$:

$$v_1 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \tag{3}$$

The general solution is $x(t) = c_1 e^t v_1 + c_2 e^{2t} v_2 + c_3 e^{3t} v_3$. Applying the initial condition:

$$\begin{pmatrix} 0 & 1 & 2 \\ -2 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \implies c_1 = 1, c_2 = 2, c_3 = 0 \tag{4}$$

The unique solution is $x(t) = e^t \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} + 2e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Behavior as $t \to \infty$: The components are governed by e^{2t} and e^t. Both terms diverge to infinity, so $\lim_{t\to\infty} \|x(t)\| = \infty$.

(ii) For the system $x' = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} x$ with $x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$: The eigenvalues are $\lambda = -1 \pm i$. For $\lambda = -1 + i$, the eigenvector is $v = \begin{pmatrix} 2+i \\ 1 \end{pmatrix}$. The complex solution is $e^{(-1+i)t} \begin{pmatrix} 2+i \\ 1 \end{pmatrix} = e^{-t}(\cos t + i \sin t) \begin{pmatrix} 2+i \\ 1 \end{pmatrix}$. Taking the real and imaginary parts produces the general solution:

$$x(t) = c_1 e^{-t} \begin{pmatrix} 2\cos t - \sin t \\ \cos t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \cos t + 2\sin t \\ \sin t \end{pmatrix} \tag{5}$$

Applying $x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ yields $c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \implies c_1 = 1, c_2 = -1$. The unique solution is $x(t) = e^{-t} \begin{pmatrix} \cos t - 3\sin t \\ \cos t - \sin t \end{pmatrix}$. Behavior as $t \to \infty$: Due to the exponential decay factor e^{-t}, $\lim_{t\to\infty} x(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The origin is a stable spiral.

3. General Solution in Real-Valued Functions

For $x' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} x$: The characteristic polynomial is $\lambda^3 + 4\lambda^2 + 7\lambda + 6 = 0$. The roots are $\lambda_1 = -2$ and $\lambda = -1 \pm i\sqrt{2}$. For $\lambda_1 = -2$, the real eigenvector is $v_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$. For $\lambda = -1 + i\sqrt{2}$, an eigenvector is $v = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + i \begin{pmatrix} \sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix}$. Taking the real and imaginary parts of $e^{(-1+i\sqrt{2})t} v$:

- Real part: $e^{-t}\begin{pmatrix} -\sqrt{2}\sin(\sqrt{2}t) \\ \cos(\sqrt{2}t) \\ -\cos(\sqrt{2}t) - \sqrt{2}\sin(\sqrt{2}t) \end{pmatrix}$.
- Imaginary part: $e^{-t}\begin{pmatrix} \sqrt{2}\cos(\sqrt{2}t) \\ \sin(\sqrt{2}t) \\ -\sin(\sqrt{2}t) + \sqrt{2}\cos(\sqrt{2}t) \end{pmatrix}$.

The general solution in terms of real-valued functions is:

$$x(t) = c_1 e^{-2t}\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} -\sqrt{2}\sin(\sqrt{2}t) \\ \cos(\sqrt{2}t) \\ -\cos(\sqrt{2}t) - \sqrt{2}\sin(\sqrt{2}t) \end{pmatrix} + c_3 e^{-t}\begin{pmatrix} \sqrt{2}\cos(\sqrt{2}t) \\ \sin(\sqrt{2}t) \\ -\sin(\sqrt{2}t) + \sqrt{2}\cos(\sqrt{2}t) \end{pmatrix} \tag{6}$$

4. Fundamental Matrices

For the system in Question 1, $x' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix} x$: Using the independent solutions, a fundamental matrix $\Psi(t)$ is:

$$\Psi(t) = \begin{pmatrix} 3e^{-t} & 0 & -4e^{-2t} \\ -4e^{-t} & e^{2t} & 5e^{-2t} \\ -2e^{-t} & -e^{2t} & 7e^{-2t} \end{pmatrix} \tag{7}$$

To find $\Phi(t)$ satisfying $\Phi(0) = I$, we compute $\Phi(t) = \Psi(t)\Psi^{-1}(0)$.

$$\Psi(0) = \begin{pmatrix} 3 & 0 & -4 \\ -4 & 1 & 5 \\ -2 & -1 & 7 \end{pmatrix} \implies \Psi^{-1}(0) = \frac{1}{12}\begin{pmatrix} 12 & 4 & 4 \\ 18 & 13 & 1 \\ 6 & 3 & 3 \end{pmatrix} \tag{8}$$

Multiplying $\Psi(t)\Psi^{-1}(0)$ produces the unique fundamental matrix:

$$\Phi(t) = \frac{1}{12}\begin{pmatrix} 36e^{-t} - 24e^{-2t} & 12e^{-t} - 12e^{-2t} & 12e^{-t} - 12e^{-2t} \\ -48e^{-t} + 18e^{2t} + 30e^{-2t} & -16e^{-t} + 13e^{2t} + 15e^{-2t} & -16e^{-t} + e^{2t} + 15e^{-2t} \\ -24e^{-t} - 18e^{2t} + 42e^{-2t} & -8e^{-t} - 13e^{2t} + 21e^{-2t} & -8e^{-t} - e^{2t} + 21e^{-2t} \end{pmatrix} \tag{9}$$

5. Harmonic Oscillator System

For $u'' + \omega^2 u = 0$ with $u(0) = u_0$ and $u'(0) = v_0$:

(i) Let $x_1 = u, x_2 = u'$. The equivalent first-order system is:

$$x' = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix} \tag{10}$$

(ii) Using the Taylor series $\exp(At) = \sum_{n=0}^{\infty} \frac{(At)^n}{n!}$: Notice $A^2 = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}^2 = \begin{pmatrix} -\omega^2 & 0 \\ 0 & -\omega^2 \end{pmatrix} = -\omega^2 I$. By induction, $A^{2m} = (-1)^m \omega^{2m} I$ and $A^{2m+1} = (-1)^m \omega^{2m} A$. Separating the series into even and odd terms:

$$\exp(At) = \sum_{m=0}^{\infty} \frac{(-1)^m (\omega t)^{2m}}{(2m)!} I + \frac{1}{\omega} \sum_{m=0}^{\infty} \frac{(-1)^m (\omega t)^{2m+1}}{(2m+1)!} A \tag{11}$$

Recognizing the Maclaurin series for sine and cosine completes the proof:

$$\exp(At) = I\cos(\omega t) + A\frac{\sin(\omega t)}{\omega} \tag{12}$$

(iii) The solution is $x(t) = \exp(At)x(0)$:

$$x(t) = \begin{pmatrix} \cos(\omega t) & \frac{1}{\omega}\sin(\omega t) \\ -\omega\sin(\omega t) & \cos(\omega t) \end{pmatrix} \begin{pmatrix} u_0 \\ v_0 \end{pmatrix} = \begin{pmatrix} u_0\cos(\omega t) + \frac{v_0}{\omega}\sin(\omega t) \\ -u_0\omega\sin(\omega t) + v_0\cos(\omega t) \end{pmatrix} \tag{13}$$

6. Repeated Eigenvalues

For $x' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} x$: The characteristic equation is $-(\lambda+1)(\lambda-2)^2 = 0$. For $\lambda = -1$, the eigenvector is $v_1 = \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix}$. For the repeated root $\lambda = 2$, there is only one independent eigenvector $v_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$. To find the generalized eigenvector w, solve $(A - 2I)w = v_2$:

$$\begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \tag{14}$$

A valid solution is $w = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$. The general solution is:

$$x(t) = c_1e^{-t}\begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix} + c_2e^{2t}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} + c_3e^{2t}\left[\begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} + t\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}\right] \tag{15}$$

7. Lower Triangular System IVP

For $x' = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 3 & 6 & 2 \end{pmatrix} x$ with $x(0) = \begin{pmatrix} -1 \\ 2 \\ -30 \end{pmatrix}$: Since the matrix is lower triangular, the eigenvalues are simply the diagonal entries: $\lambda_1 = 1$ (multiplicity 2) and $\lambda_2 = 2$. For $\lambda_2 = 2$, the eigenvector is $v_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. For $\lambda_1 = 1$, the eigenvector is $v_2 = \begin{pmatrix} 0 \\ 1 \\ -6 \end{pmatrix}$, and the generalized eigenvector from $(A - I)w = v_2$ is $w = \begin{pmatrix} -1/4 \\ 0 \\ -21/4 \end{pmatrix}$. The general solution is $x(t) = c_1e^{2t}v_1 + c_2e^tv_2 + c_3e^t(w + tv_2)$. Applying the initial condition $x(0) = c_1v_1 + c_2v_2 + c_3w$ evaluates to $c_1 = 3, c_2 = 2, c_3 = 4$. The particular solution is:

$$x(t) = \begin{pmatrix} -e^t \\ 2e^t + 4te^t \\ 3e^{2t} - 33e^t - 24te^t \end{pmatrix} \tag{16}$$

8. Matrix Exponential for a Jordan Block

Let $J = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$. Note that $J = \lambda I + N$ where $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, and $N^3 = 0$.

(i) Powers of J:

- $J^2 = \begin{pmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{pmatrix}$.
- $J^3 = \begin{pmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{pmatrix}$.
- $J^4 = \begin{pmatrix} \lambda^4 & 4\lambda^3 & 6\lambda^2 \\ 0 & \lambda^4 & 4\lambda^3 \\ 0 & 0 & \lambda^4 \end{pmatrix}$.

(ii) Inductive argument for J^n: Base Case: $n = 1$ yields $J^1 = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$, which matches the formula. Inductive Step: Assume the formula holds for $n = k$. We compute $J^{k+1} = J \cdot J^k$:

$$\begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix} \begin{pmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{pmatrix} = \begin{pmatrix} \lambda^{k+1} & (k+1)\lambda^k & (\frac{k^2-k}{2} + k)\lambda^{k-1} \\ 0 & \lambda^{k+1} & (k+1)\lambda^k \\ 0 & 0 & \lambda^{k+1} \end{pmatrix} \tag{17}$$

Since $\frac{k^2-k}{2} + k = \frac{k(k+1)}{2}$, the property holds for $k+1$. The proof is complete.

(iii) To determine $\exp(Jt)$: Since (λtI) and (tN) commute, $\exp(Jt) = \exp(\lambda tI)\exp(tN)$. Since $N^k = 0$ for $k \geq 3$:

$$\exp(Jt) = e^{\lambda t}\left(I + tN + \frac{t^2}{2}N^2\right) = e^{\lambda t}\begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} \tag{18}$$

9. Nonhomogeneous Systems

We use Variation of Parameters: $x(t) = \Psi(t)c + \Psi(t)\int \Psi^{-1}(s)g(s)ds$.

(i) For $x' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x + \begin{pmatrix} e^t \\ t \end{pmatrix}$: The fundamental matrix for the homogeneous system is $\Psi(t) = \begin{pmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{pmatrix}$. Applying the variation of parameters integral evaluates to the particular solution $x_p(t) = \begin{pmatrix} \frac{3}{2}te^t + t - \frac{1}{4}e^t \\ \frac{3}{2}te^t + 2t - \frac{3}{4}e^t - 1 \end{pmatrix}$. General solution:

$$x(t) = c_1 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} \frac{3}{2}te^t + t - \frac{1}{4}e^t \\ \frac{3}{2}te^t + 2t - \frac{3}{4}e^t - 1 \end{pmatrix} \tag{19}$$

(ii) For $x' = \begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix} x + \begin{pmatrix} e^t \\ \sqrt{3}e^{-t} \end{pmatrix}$: The fundamental matrix is $\Psi(t) = \begin{pmatrix} \sqrt{3}e^{2t} & e^{-2t} \\ e^{2t} & -\sqrt{3}e^{-2t} \end{pmatrix}$. Integration yields the particular solution $x_p(t) = \begin{pmatrix} -\frac{2}{3}e^t - e^{-t} \\ -\frac{\sqrt{3}}{3}e^t + \frac{2\sqrt{3}}{3}e^{-t} \end{pmatrix}$. General solution:

$$x(t) = c_1 e^{2t} \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix} + \begin{pmatrix} -\frac{2}{3}e^t - e^{-t} \\ -\frac{\sqrt{3}}{3}e^t + \frac{2\sqrt{3}}{3}e^{-t} \end{pmatrix} \tag{20}$$

(iii) For $x' = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix} x + \begin{pmatrix} \csc t \\ \sec t \end{pmatrix}$ on $\frac{\pi}{2} < t < \pi$: The fundamental matrix is $\Psi(t) = \begin{pmatrix} 2\cos t - \sin t & \cos t + 2\sin t \\ \cos t & \sin t \end{pmatrix}$. Calculating $\Psi^{-1} g$ produces $\begin{pmatrix} 2\tan t \\ \cot t + \tan t - 2 \end{pmatrix}$. Integrating on the specified interval gives particular coefficients $u_1(t) = -2\ln(-\cos t)$ and $u_2(t) = \ln(-\tan t) - 2t$. General solution:

$$x(t) = c_1 \begin{pmatrix} 2\cos t - \sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} \cos t + 2\sin t \\ \sin t \end{pmatrix} + \begin{pmatrix} (2\cos t - \sin t)(-2\ln(-\cos t)) + (\cos t + 2\sin t)(\ln(-\tan t) - \\ -2\cos t \ln(-\cos t) + \sin t(\ln(-\tan t) - 2t) \end{pmatrix} \tag{21}$$